

OPTIMIZING EMERGENCY RESPONSE LATENCY IN HIGH-RISE BUILDING FIRES VIA YOLO-BASED OBJECT DETECTION

By

Kishan

Kannan

1. Greenwood High International School, Sarjapur - Varthur, road, adjacent to ICSE Campus, Chikkavaderapura, Karnataka 560087, India.

Email Address – kishankannan.blr@gmail.com

Abstract:

High-rise and complex building fires claim numerous lives each year, and initial responders often lack immediate, organized intelligence regarding the location of victims and active hazards. This project presents a software minimum viable product (MVP) designed to optimize search-and-rescue triage and situational awareness in structural firefighting environments. By utilizing a YOLO-based object detection model, the system analyzes images from interior rooms to accurately identify humans, fire, and smoke. Rather than outputting raw, uncontextualized confidence scores, the software processes this visual evidence into a tactical triage dashboard that ranks rooms based on logical states and victim volume. These classifications directly align with emergency dispatch protocols, prioritizing rooms from critical multi-victim scenarios (life threat combined with active fire) down to lower-tier suppression zones (active fire with no visible victims). Ultimately, this system does not make autonomous rescue decisions; instead, it empowers incident command with organized, actionable data, reducing emergency response latency and allowing firefighters to rapidly direct Search & Rescue (SAR) and Suppression teams where they are needed most.

Keywords:

Keywords: Computer Science and Software Engineering; Artificial Intelligence; Computer Vision; Human Detection in Smoke; High-Rise Building Fire Rescue; Autonomous UAV

Introduction:

The integration of artificial intelligence (AI) and computer vision is transforming fire rescue operations, particularly in hazardous high-rise building scenarios where immediate situational awareness is heavily restricted. During structural fires, low visibility due to smoke, heat, and structural instability creates massive challenges for incident commanders trying to locate trapped individuals. Recent research demonstrates that deep learning models—specifically YOLO variants and

Convolutional Neural Networks (CNNs)—can rapidly identify and localize humans in these dangerous environments by processing interior imagery. Ongoing research highlights how these models operate effectively under fire conditions. For instance, the study “*Detection of Human Life in Fire Scenarios Using YOLOv8*” demonstrates the model's ability to utilize OpenCV frameworks to successfully isolate human positions within bounding boxes, even in complex scenes. While this paper provides a robust computational subsystem, its findings yield raw detection data (such as confidence scores) rather than actionable tactical intelligence. In a life-or-death scenario, raw numbers are insufficient for emergency dispatchers who must immediately decide where to send specific teams. This project bridges that gap by proposing a software Minimum Viable Product (MVP) focused on realistic decision support rather than autonomous robotics. By analyzing images from various rooms to detect humans, fire, and smoke, the system synthesizes visual evidence into an action-oriented triage dashboard. Instead of making the final rescue decision, the software organizes operational intelligence by ranking rooms based on rescue urgency and logical states—prioritizing multi-victim critical zones requiring immediate Search & Rescue (SAR) over active fire zones requiring Suppression teams. This approach synthesizes the latest advancements in AI detection models with practical, human-in-the-loop disaster response protocols.